Unit 22

Monopolistic Competition

After completing this unit, the student will be able to:

-Understand the features of Monopolistic Competition

-Analyse the Competitive Equilibrium of Monopolistic Competition

Monopolistic competition

The model of monopolistic competition describes a common market structure in which firms have many competitors, but each one sells a slightly different product. Monopolistic competition as a market structure was first identified in the 1930s by American economist Edward Chamberlin, and English economist Joan Robinson. Many small businesses operate under conditions of monopolistic competition, including independently owned and operated high-street stores and restaurants. In the case of restaurants, each one offers something different and possesses an element of uniqueness, but all are essentially competing for the same customers.

Characteristics

Monopolistically competitive markets exhibit the following characteristics:

Large number of firms:

There are usually a large number of independent firms competing in the market.

Independent Action by firms:

Each firm makes independent decisions about price and output, based on its product, its market, and its costs of production.

Imperfect but wide spread knowledge:

Knowledge is widely spread between participants, but it is unlikely to be perfect.

More roles to the producer:

The entrepreneur has a more significant role than in firms that are perfectly competitive because of the increased risks associated with decision making.

Existence of freedom to entry and exit:

There is freedom to enter or leave the market, as there are no major barriers to entry or exit.

Product differentiation:

A central feature of monopolistic competition is that products are differentiated. There are four main types of differentiation:

-Physical product differentiation, where firms use size, design, colour, shape, performance, and features to make their products different. For example, consumer electronics can easily be physically differentiated.

-Marketing differentiation, where firms try to differentiate their product by distinctive packaging and other promotional techniques. For example, breakfast cereals can easily be differentiated through packaging. Human capital differentiation, where the firm creates differences through the skill of its employees, the level of training received, distinctive uniforms, and so on.

-Differentiation through distribution, including distribution via mail order or through internet shopping, such as Amazon.com, which differentiates itself from traditional bookstores by selling online.

Firms are facing down ward slopping demand curve:

Firms are price makers and are faced with a downward sloping demand curve. Because each firm makes a unique product, it can charge a higher or lower price than its rivals. The firm can set its own price and does not have to ‘take' it from the industry as a whole, though the industry price may be a guideline, or becomes a constraint. This also means that the demand curve will slope downwards.

Existence of selling cost:

Firms operating under monopolistic competition usually have to engage in advertising. Firms are engaged in competition with other firms offering a similar product or service, and may need to advertise to let customers know their differences. Common methods of advertising for these firms are through local press and radio, local cinema, posters, leaflets and special promotions.

Objective of profit maximisation:

Monopolistically competitive firms are assumed to be profit maximisers because firms tend to be small with entrepreneurs actively involved in managing the business.

Monopolistic competition in the short run:

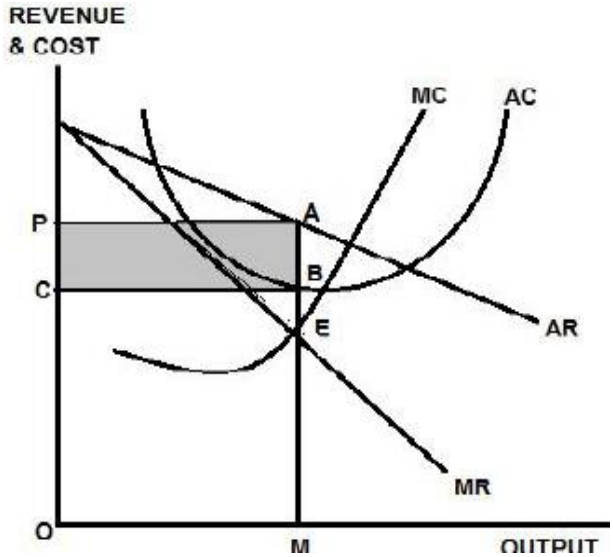
I. **Abnormal Profit**

In the short run supernormal profits are possible, but in the long run new firms are attracted into the industry, because of low barriers to entry, good knowledge and an opportunity to differentiate. Two important conditions are to be satisfied to attain short run equilibrium under monopolistically competitive firms. They are:

1. Marginal Cost = Marginal Revenue (MC=MR)
2. MC should Cut MR curve from below

Such a condition is explained in the figure 22.1

Figure 22.1



At profit maximisation, equilibrium point is attained at point E where, MC = MR, and the firm is producing output OM and price is OP. At this stage:

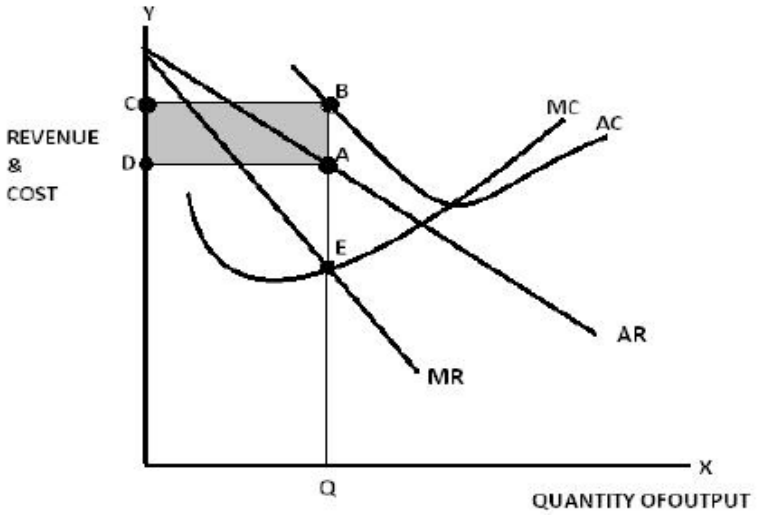
- AR = MA AC= MB
- Since AR > AC, the firm is gaining abnormal profit
- Profit per unit = MA – MB = AB
- Total abnormal profit of the firm at OQ output = OM x AB = PABC (Shaded area).

As new firms enter the market, demand for the existing firm's products becomes more elastic and the demand curve shifts to the left, driving down price. Eventually, all supernormal profits are eroded away.

II. Loss Position

In the short run, there is a possibility of loss also to a firm under monopolistic competition. Such a case is demonstrated in the figure 22.2.

Figure 22.2



The equilibrium point is attained at point E where, MC = MR, and the firm is producing output OQ and price is OC. At this stage:

- AR = QA
- AC= QB
- Since AR < AC, the firm is suffering loss.
- Loss per unit = QB – QA = AB
- Total loss of the firm at OQ output = OQ x AB = ABCD (the shaded area)

As loss suffering firms leave the market, demand for the existing firm's products becomes less elastic and the demand curve shifts to the right, which push the price up. Eventually, all loss will be covered and the firm starts to earn normal profit in the long run.

Monopolistic competition in the long run

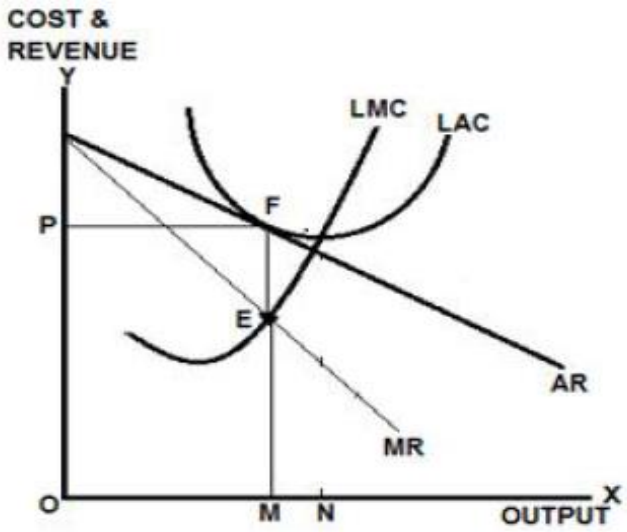
The freedom to enter and to leave the industry in a monopolistically competitive market will lead to changes in the supply of the products in the market. Abnormal profit of the existing firm in the short run will attract new firms to the industry. Similarly, firms suffering with loss will leave the industry. Therefore, in the long run all the existing firms will earn only normal profit.

Three conditions are to be satisfied for long run equilibrium under MPC.

1. Marginal Cost= Marginal Revenue (MC=MR)
2. MC should Cut MR curve from below
3. Average Revenue should be equal to Average Total cost (AR=ATC)

The normal profit of firm in the long run is demonstrated in the figure 22.3.

Figure 22.3



The equilibrium point is attained at point E where MC = MR, and the firm is producing output OM and price is OP. At this stage:

- AR = MF
- AC= MF
- Since AR = AC, the firm is earning only normal profit

Hence the three conditions of equilibrium for the long run are satisfied. Since the firm is earning normal profit, it has no tendency to change its scale of operation.

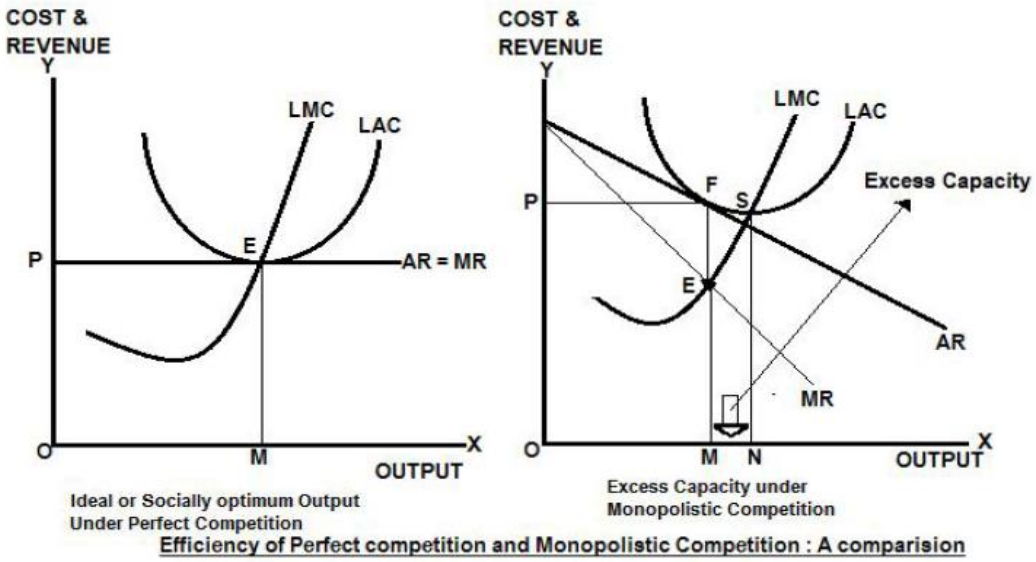
Excess capacity

Firms under monopolistic competition restrict output in order to maintain higher prices than firms in competitive markets. Firms under monopolistic competition operate at higher average total cost than equivalent firms in competitive markets. This implies that not all production efficiencies are being captured in monopolistically competitive markets. The amount by which the average-cost-minimizing quantity exceeds the quantity produced by a firm under monopolistic competition is referred to as “excess capacity.”

Excess Capacity = Optimum production capacity - Equilibrium output under MP

The firm is allocatively and productively inefficient in both the long and short run. Consider the comparative analysis of long run equilibrium under perfect competition and monopolistic competition.

Figure 22.4



Efficiency of Perfect competition and Monopolistic Competition : A comparision

In the figure 22.4, Ideal output is produced by perfect competitive firm because the firm is producing at the minimum of its long run average cost. It means that the firm is using its optimum capacity. But under monopolistic competitive market, the firm is operating before the optimum because they are forced to stop production when average cost is decreasing. So, the cost and prices are higher in monopolistically competitive market. Here MN level of output cannot be produced by the firm and this unproduced output is termed as excess capacity. This means they are productively inefficient in both the long and short run. As an economic model of competition, monopolistic competition is more realistic than perfect competition - many familiar and commonplace markets have many of the characteristics of this model.

Selling Costs

Under monopolistic competition, products are differentiated and these differences are made known to the buyers through selling costs. Selling costs refer to the expenses incurred on marketing, sales promotion and advertisement of the product. Such costs are incurred to persuade the buyers to buy a particular brand of the product in preference to competitor's brand. Due to this reason, selling costs constitute a substantial part of the total cost under monopolistic competition.

The selling costs, according to Chamberlin, include "advertising in its many forms, salaries of salesmen and the expenses of sales departments and sales agencies (except where these agencies actually handle the goods), window displays, and displays and demonstration of all kinds." It must be noted that there are no selling costs in perfect competition as there is perfect

knowledge among buyers and sellers. Similarly, under monopoly, selling costs are of small amount (only for informative purpose) as the firm does not face competition from any other firm.

Fixed Costs and the Number of Firms

The number of firms in a monopolistically competitive equilibrium depends on firms' costs. The larger each firm's fixed cost, the smaller the number of monopolistically competitive firms in the market equilibrium.

Although entry is free, if the fixed costs are high, few firms may enter. In the auto mobile industry, just to develop a new fender costs $8 to $10 million. Developing a new pharmaceutical drug may cost $350 million or more.

We can illustrate this relationship using the airlines example, in which we modify our assumptions about entry and fixed costs. American and United are the only airlines providing service on the Chicago–Los Angeles route. Until now, we have assumed that a barrier to entry—such as an inability to obtain landing rights at both airports—prevented entry and that the firms had no fixed costs. If fixed cost is zero and marginal cost is constant at $147 per passenger, average cost is also constant at $147 per passenger. As we showed earlier, each firm in this oligopolistic market flies q = 64 per quarter at a price of p = $211 and makes a profit of $4.1 million per quarter.

Now suppose that the market has no barriers to entry, but each airline incurs a fixed cost to enter of F due to airport fees, capital expenditure, or other factors. Each firm's marginal cost remains $147 per passenger, but its average cost,

$$AC = 147 + \frac{F}{q}$$

falls as the number of passengers rises, as panels a and b of Figure 22.5 illustrate for F = $2.3 million.

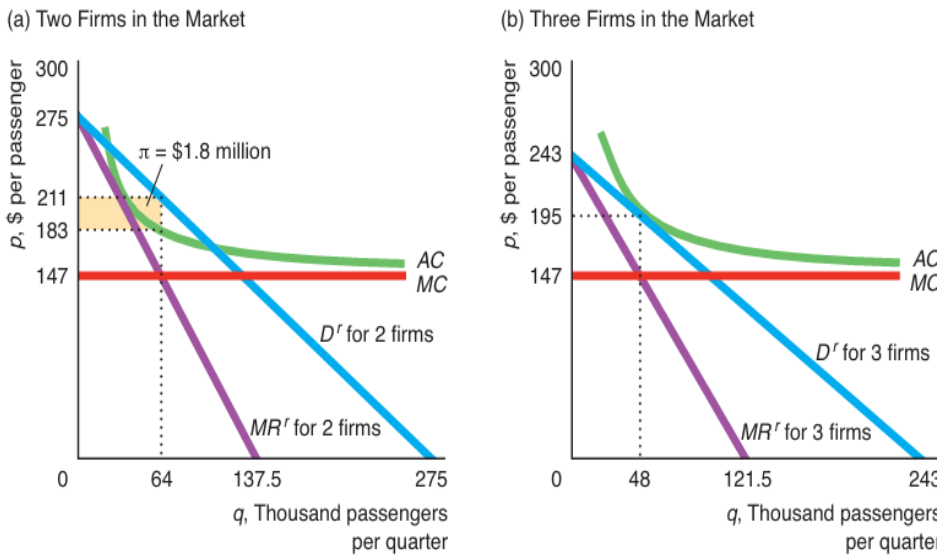


Figure 22.5: Monopolistic Competition Among Airlines

If the monopolistically competitive market has only two firms the fixed costs must be $4.1 million per firm. Because we know that these firms receive a profit of $4.1 million per firm in the absence of fixed costs. With this fixed cost, the monopolistically competitive price and quantity are the same as in the oligopolistic equilibrium, q = 64 and p = $211, and the number of firms is the same, but now each firm's profit is zero.

If the fixed cost is only $2.3 million and the market has only two firms, each firm makes a profit, as panel a show. Each duopoly firm faces a residual demand curve (labelled "D^r for 2 firms"), which is the market demand minus its rival's Nash Cournot equilibrium quantity, q = 64. Given this residual demand, each firm produces q = 64, which equates its marginal revenue, MR^r, and its marginal cost, MC. At q = 64, the firm's average cost is AC = $147 + ($2.3 million) / (64 units) ≈ $183, so each firm makes a profit of $\pi = (p - AC)\, q \approx (\$211 - \$183) \times 64$ units per quarter ≈ $1.8 million per quarter.

This substantial economic profit attracts an entrant. The entry of a third firm causes the residual demand for any one firm to shift to the left in panel b. In the new equilibrium, each firm sets q = 48 and charges p = $195. At this quantity, each firm's average cost is $195, so the firms break even. No other firms enter because if one did, the residual demand curve would shift even farther to the left and all the firms would lose money. Thus, if fixed cost is $2.3 million, the market has three firms in the monopolistically competitive equilibrium. This example illustrates a general result: The lower the fixed costs, the more firms in the monopolistically competitive equilibrium.

Multiple Choice Questions

1) Which of the following is a characteristic of monopolistic competition?
    a) Firms have no control over pricing.
    b) There is a large number of firms selling differentiated products.
    c) All firms produce identical products.
    d) There are significant barriers to entry and exit.

Answer: B

2) In monopolistic competition, firms are considered price makers because:
    a) They can set their prices based on the industry price.
    b) They face a perfectly elastic demand curve.
    c) They sell differentiated products that allow them to charge different prices.
    d) They have no competitors in the market.

Answer: C

3) What happens to the demand curve of a monopolistically competitive firm when new firms enter the market?
    a) It becomes more inelastic.
    b) It shifts to the right.
    c) It shifts to the left, making demand more elastic.
    d) It remains unchanged.

Answer: C

4) Under monopolistic competition, an increase in selling costs primarily affects which of the following?
    a) The fixed costs of the firm.
    b) The price and competitiveness of the product.
    c) The marginal cost of production.
    d) The number of firms in the market.

Answer: B

5) Which of the following best describes the long-run equilibrium for firms in monopolistic competition?

a) Firms earn supernormal profits indefinitely.

b) Firms will minimize costs by operating at their optimal capacity.

c) Firms earn zero economic profits as new entrants drive prices down.

d) Firms will always face increasing average costs.

Answer: C